Suleyman Talha Belen

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Education

B.Sc. in Electrical and Electronics Engineering 09/2023 – 06/2027
Bilkent University · CGPA: 3.5/4.0 · Full Merit Scholarship
Relevant Coursework: Learning for Robotics, Computer Vision, Signals and Systems, Nonlinear Optimization

Visiting Student (Summer Session), Computer Science 2025
Stanford University
CS229 (Machine Learning), EE364A (Convex Optimization)

Visiting Student, EECS and Engineering 2024
University of California, Berkeley
EECS 16A (Designing Information Devices and Systems), ENG 170 (Technology Leadership)

Research

New York University – Research Intern 01/2026 – Present
Advised by: Lerrel Pinto. World models for dexterous manipulation learned from human video, and physics-based retargeting of human demonstrations to robot hands, deployed in simulation and on real robots.

Stanford University, CVXPY – Remote 12/2025 – Present
Post-solver feasibility and KKT optimality checks, including Disciplined Nonlinear Programming (DNLP) problems.

Bilkent University, LIRA Lab 07/2026 – Present
Advised by: Özgür S. Oğuz. Obtaining physically reliable robot data from RGB cameras without costly hardware.

Industry Collaborations

Bin Picking for Industrial Manipulation – Bilkent × ETİ Makine 09/2026 – Present
Robotic bin picking of cluttered, complex objects under harsh industrial conditions using a Zivid 3D camera.

Construction Robotics – Karakol (Italy) 06/2026 – Present
Helping an architecture practice move toward robotic house building; interviewed teams in Australia, Italy and China, and now collaborating with Karakol on modular construction robotics.

Teaching

Study Session Instructor, Stanford University – EE364A 2025
Co-led review sessions on key convex optimization concepts for the full EE364A class.

Employment

AI Systems Test Engineer Intern, Havelsan 06/01/2026 – 06/26/2026
Tested edge cases of AI-based computer vision models deployed on city surveillance cameras, evaluating performance on traffic accident detection and fire/smoke classification under diverse real-world conditions.

Applied Research Intern, SDT 08/2025 – 09/2025
Conceptual research on seismic data visualization inspired by Neural Radiance Fields for Turkish Petroleum to determine the shape of underwater rocks.

Honors and Awards

Full Merit Scholarship, Bilkent University (including monthly stipend) 2022 – 2027
UC Berkeley International Study Scholarship, Recipient of $2,000 2026
YKS University Entrance Exam, 442nd nationwide among ∼2 million students, quantitative track 2022

Selected Projects

1. **S. T. Belen.** *"Safe Swarm Control with Higher-Order CBFs."* Code.
2. **S. T. Belen.** *"PPO-MAML for Meta-World Assembly-v3."* Code.
3. **S. T. Belen.** *"Domain-Invariant Racer (CarRacing-v3)."* Code.
4. **S. T. Belen.** *"Multimodal Audio Tokenizer."* Code.
5. **S. T. Belen.** *"SwinIR Super-Resolution."* Technical Report.
6. **S. T. Belen et al.** *"ToolMind: VLM-Guided Spatial Reasoning for Robotic Tool Use"* **Best Project Award.** Project Page.

Activities

Team Lead, BilkentSAT – Student Satellite Team 2024 – 2025
Stanford–Bilkent Exchange: organized Zoom sessions connecting Stanford students with leaders of Bilkent's Medeniyet society
Stanford Robotics Seminars, Attendee
Member, Bilkent AI

Skills

Dexterous Manipulation and Robot Learning (MuJoCo, retargeting, sampling-based MPC, RL, behavior cloning)
Robot Hardware (Wuji hand, Franka, xArm, YOR, Zivid 3D camera)
Optimization (CVXPY, convex and nonlinear optimization, Control Barrier Functions)
Machine Learning and Signal Processing (PyTorch, scikit-learn, NumPy, Pandas, MATLAB)
Programming Languages: Python, MATLAB, C, VHDL, Assembly

Spoken & Written Languages

Turkish Native Proficiency
English *(TOEFL iBT: 110/120, CEFR C1, 2026)* Full Professional Proficiency